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Experiment: Tic-Tac-Toe Game

Aim

To implement the **Tic-Tac-Toe Game** using Python, allowing two players to play on a 3×3 game board.

Objectives

- To understand the implementation of the Tic-Tac-Toe game.
- To develop a Python program for a two-player game.
- To check for winning conditions after each move.
- To detect a draw when no moves are left.
- To understand the application of decision-making in game development.

Algorithm

1. Create a 3×3 game board.
2. Display the board to the players.
3. Allow players **X** and **O** to enter their moves alternately.
4. Validate the entered position.
5. Check for a winner after every move.
6. If all cells are filled without a winner, declare the game a draw.
7. Display the final result.

Python Program

```
# Tic-Tac-Toe Game


board = [' ' for _ in range(9)]


def display():
```

```
    print("\n")
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])
    print()

def winner(player):
    win = [
        [0,1,2], [3,4,5], [6,7,8],
        [0,3,6], [1,4,7], [2,5,8],
        [0,4,8], [2,4,6]
    ]

    for combo in win:
        if all(board[pos] == player for pos in combo):
            return True
    return False

player = "X"

for turn in range(9):
    display()

    move = int(input(f"Player {player}, enter position (1-9): ")) - 1
```

```
    if move < 0 or move > 8 or board[move] != ' ':
        print("Invalid Move! Try Again.")
        continue

    board[move] = player

    if winner(player):
        display()
        print("Player", player, "Wins!")
        break

    player = "O" if player == "X" else "X"

else:
    display()
    print("Game Draw!")
```

Sample Output

```
 |  |
--+---+--
 |  |
--+---+--
 |  |

Player X, enter position (1-9): 1
```

```
X |   |
--+---+--
  |   |
--+---+--
  |   |

Player O, enter position (1-9): 5

X |   |
--+---+--
  | O |
--+---+--
  |   |

Player X, enter position (1-9): 2
Player O, enter position (1-9): 8
Player X, enter position (1-9): 3

X | X | X
--+---+--
  | O |
--+---+--
  | O |

Player X Wins!
```

Out Put

```python
# Tic-Tac-Toe Game

board = [' ' for _ in range(9)]

def display():
    print("\n")
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])
    print()

def winner(player):
    win = [
        [0,1,2], [3,4,5], [6,7,8],
        [0,3,6], [1,4,7], [2,5,8],
        [0,4,8], [2,4,6]
    ]

    for combo in win:
        if all(board[pos] == player for pos in combo):
            return True
    return False

player = "X"

for turn in range(9):
    display()
```

```
  |   |
--+---+--
  |   |
--+---+--
  |   |

Player X, enter position (1-9):
```

Result

The Tic-Tac-Toe game was successfully implemented in Python. The program allowed two players to play alternately, checked for valid moves, detected the winner, and declared a draw when appropriate.

Conclusion

The experiment demonstrated the implementation of a **Tic-Tac-Toe game** using Python. The program effectively handled player input, validated moves, detected winning combinations, and identified draw conditions. This experiment illustrates the application of decision-making, conditional statements, loops, and game logic, which are fundamental concepts in Artificial Intelligence and game development.